Quick sign test for the shortcut sign

Ratio test for Eqs. (29) and (32), using real values of z

Short verdict. The stochastic shortcut matrix, where $\lambda = \beta(1-q)$ is moved from the self-loop at u to the directed edge $u \to v$, agrees with Eq. (32) only with the corrected plus/plus sign. The original minus/minus sign does not match this stochastic matrix.

I used the proposed ratio check: compute the first-passage generating function from a full-ring shortcut propagator by

$$\widetilde{F}_{n_0}(v,z) = \frac{\widetilde{S}_{n_0}(v,z)}{\widetilde{S}_v(v,z)}.$$

I then compared this value with Eq. (32) using both signs.

There is one important point in the test. If Eq. (29) is used literally as printed in the manuscript, its ratio agrees with the original minus/minus Eq. (32). However, the actual probability-conserving stochastic shortcut matrix for $u \to v$ agrees with the corrected plus/plus Eq. (32). Thus this test pushes the sign inconsistency one step earlier: Eq. (29), as printed, has the opposite defect sign if it is interpreted as moving self-loop mass from u to v.

Sanity check. For positive real $z < 1$, a first-passage generating function has non-negative coefficients and therefore cannot be negative. In the $N = 10$, $q = 2/3$, $\beta = 2/7$, $u = 5$, $v = 0$ test, the original sign gives a negative value for $n_0 = 4$, $z = 0.8$, while the stochastic matrix and the corrected sign give a positive value.

Error Summary

The grid uses two symmetric cases, including the figure parameters in the manuscript $N = 10$, $q = 2/3$, $\beta = 2/7$, $u = 5$, $v = 0$, with several starts n_0 and real values of z. Site labels here are zero-based.

Main sign check

Quantity compared over the grid	Maximum absolute error
Original Eq. (32) vs stochastic shortcut matrix	3.669746×10^{-1}
Corrected Eq. (32) vs stochastic shortcut matrix	2.775558×10^{-16}
Literal Eq. (29) ratio vs original Eq. (32)	1.318390×10^{-16}
Stochastic-sign Eq. (29) ratio vs corrected Eq. (32)	5.551115×10^{-17}

Control checks

Quantity compared over the grid	Maximum absolute error
Eq. (19) $W_{n_0}(u,z)$ vs absorbing matrix	1.043610×10^{-14}
Eq. (28) $W_u(u,z)$ vs absorbing matrix	1.687539×10^{-14}

Representative Values

n_0	z	stochastic matrix	literal Eq. (29)	original Eq. (32)	corrected Eq. (32)
2	0.50	0.044096501039	0.042961443311	0.042961443311	0.044096501039
2	0.80	0.197025316799	0.173752283343	0.173752283343	0.197025316799
2	0.95	0.532769134231	0.394627514541	0.394627514541	0.532769134231
4	0.80	0.095953324694	-0.033502923905	-0.033502923905	0.095953324694

Why This Locates Eq. (29)

For the stochastic shortcut interpretation, the transition mass $\lambda = \beta(1-q)$ is removed from the $u \to u$ self-loop and added to the $u \to v$ edge. In the resolvent, this produces the sign pattern with a $+z\lambda\{P_u(u,z) - P_v(u,z)\}$ denominator. The printed Eq. (29) uses the opposite rank-one defect sign. Therefore the printed Eq. (29) reproduces the original Eq. (32), but both differ from the direct stochastic shortcut matrix.